NeuroLearn

All-in-One AIML Learning Platform

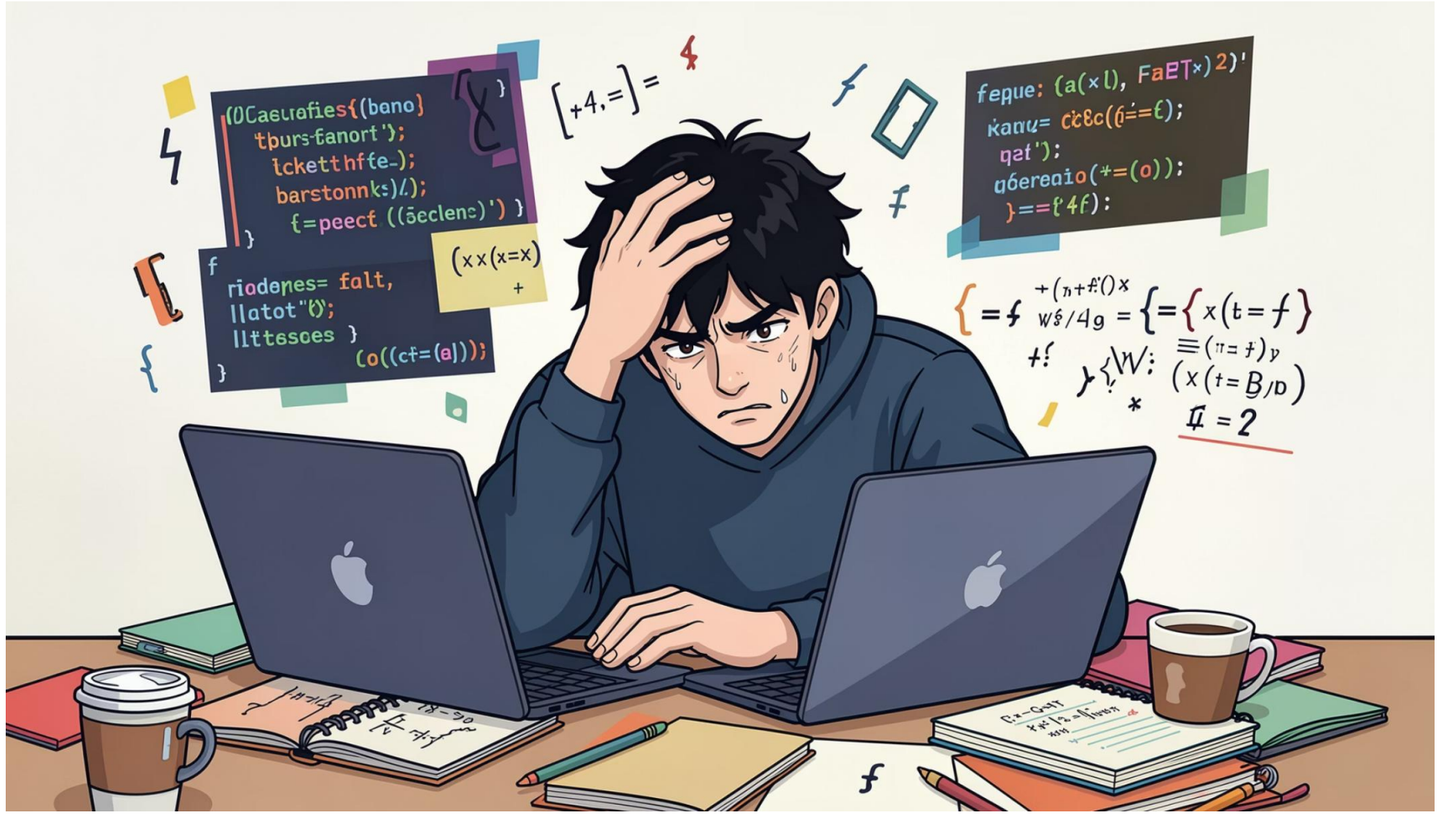
An interactive, gamified web platform that makes learning Artificial Intelligence & Machine Learning engaging, practical, and accessible — all in one place.

🏆 VIBETHON Hackathon

TEAMVISIONEERS · T143

The Problem We're Solving

AI Education Is Hard to Access — We Fixed That



Traditional AI/ML courses are expensive, fragmented, and passive. NeuroLearn brings everything into one zero-setup platform — structured learning, a real code playground, interactive games, and live simulations.

Zero Setup

Open index.html — no npm, no server, no build tools.

Fully Gamified

XP, badges, streaks, and a live leaderboard keep you hooked.

Hands-On by Design

Code, play, simulate, and quiz your way to mastery.

What's Inside NeuroLearn

Six powerful features, one seamless experience.

Learn

26 lessons across 5 tracks — Foundations to Computer Vision

Playground

Ace Editor + Skulpt Python with 8 ML templates

Games

6 interactive ML mini-games with Canvas visuals

Quiz

6 modes, 60+ questions, timed with streaks

Simulate

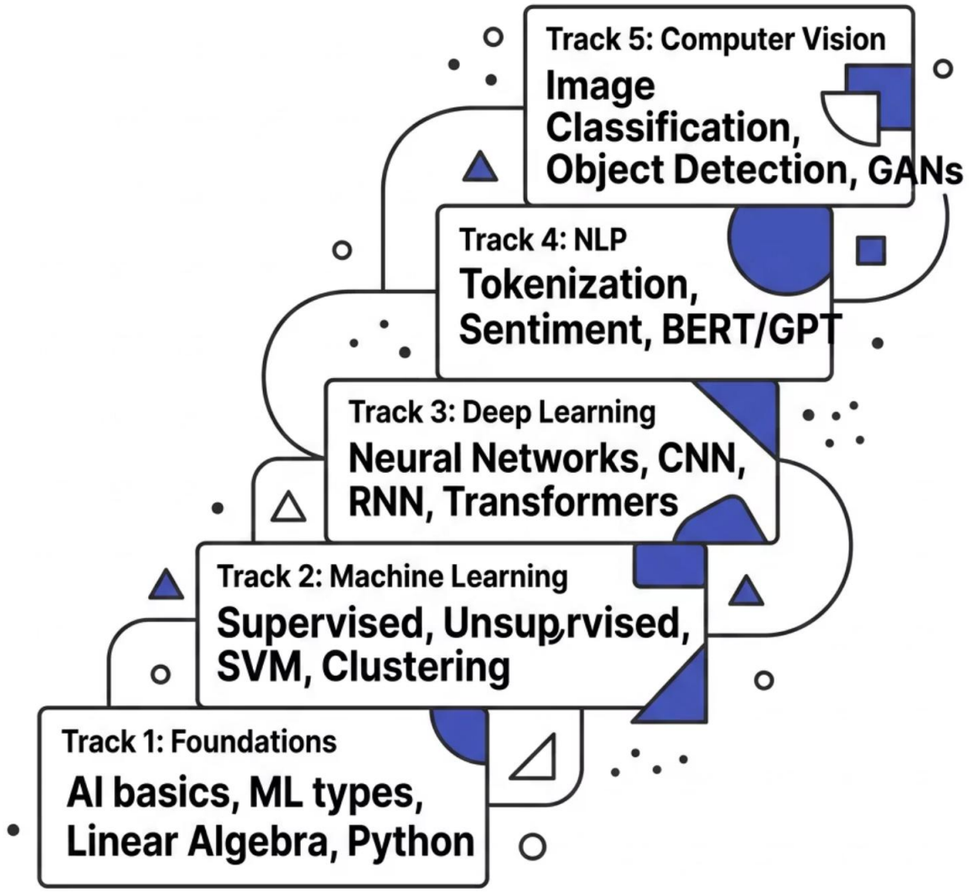
6 real-world AI demos with live datasets

NeuroBot

AI tutor with voice read-aloud & page-aware chips

📚 Structured Learning

26 Lessons Across 5 Tracks

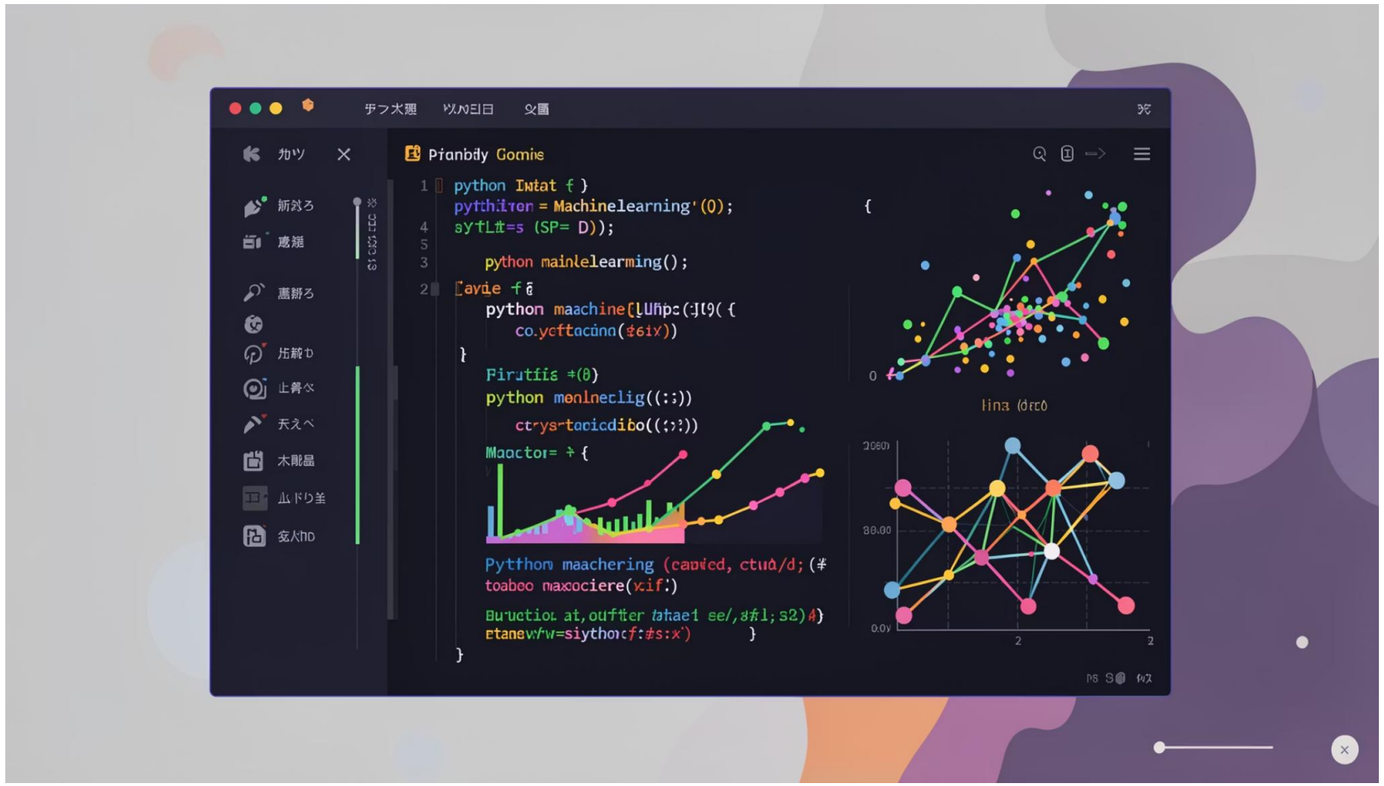


Each track builds on the last, from absolute beginner to advanced practitioner. Every lesson includes visual AI hierarchy diagrams, type cards, real-world app showcases, and an inline quiz to lock in knowledge.

ⓘ Each lesson ends with a quiz to reinforce learning before moving forward.

📟 Code Playground

Write Real Python. See Real Results.



Powered by **Ace Editor** for syntax highlighting and **Skulpt** for in-browser Python execution — no server needed. Switch between **simulated mode** (rich ML visualizations) and **execute mode** (real Python runtime).

8 ML Templates

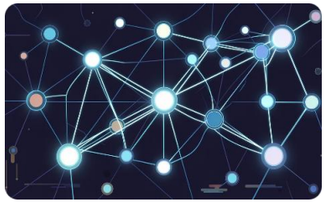
Linear Regression, KNN, K-Means, Neural Net, Decision Tree, Gradient Descent, Naive Bayes, Perceptron

Ctrl+Enter to Run

Instant Canvas visualizations — see your algorithm come alive.

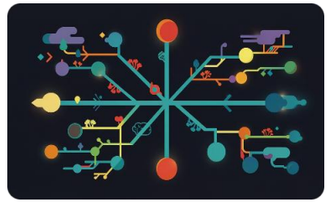
🎮 Gamification Engine

Learn by Playing — 6 Mini-Games



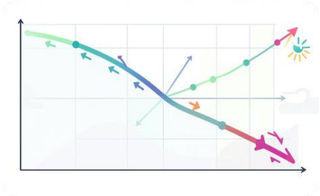
Neuron Builder

Design neural net architectures layer by layer. **+100 XP**



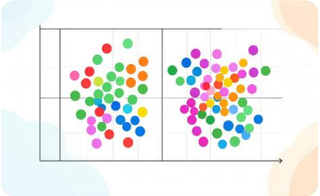
Decision Tree Explorer

Master splits and entropy visually. **+110 XP**



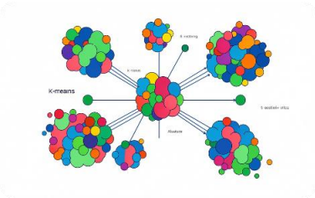
Gradient Descent ⭐

Tune learning rates and watch optimization in action. **+150 XP**



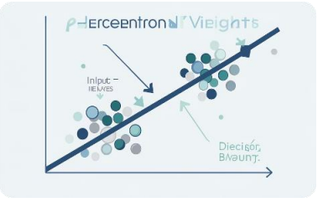
Classification Challenge

Draw decision boundaries, maximize F1 score. **+140 XP**



K-Means Adventure

Unsupervised clustering made fun and intuitive. **+130 XP**

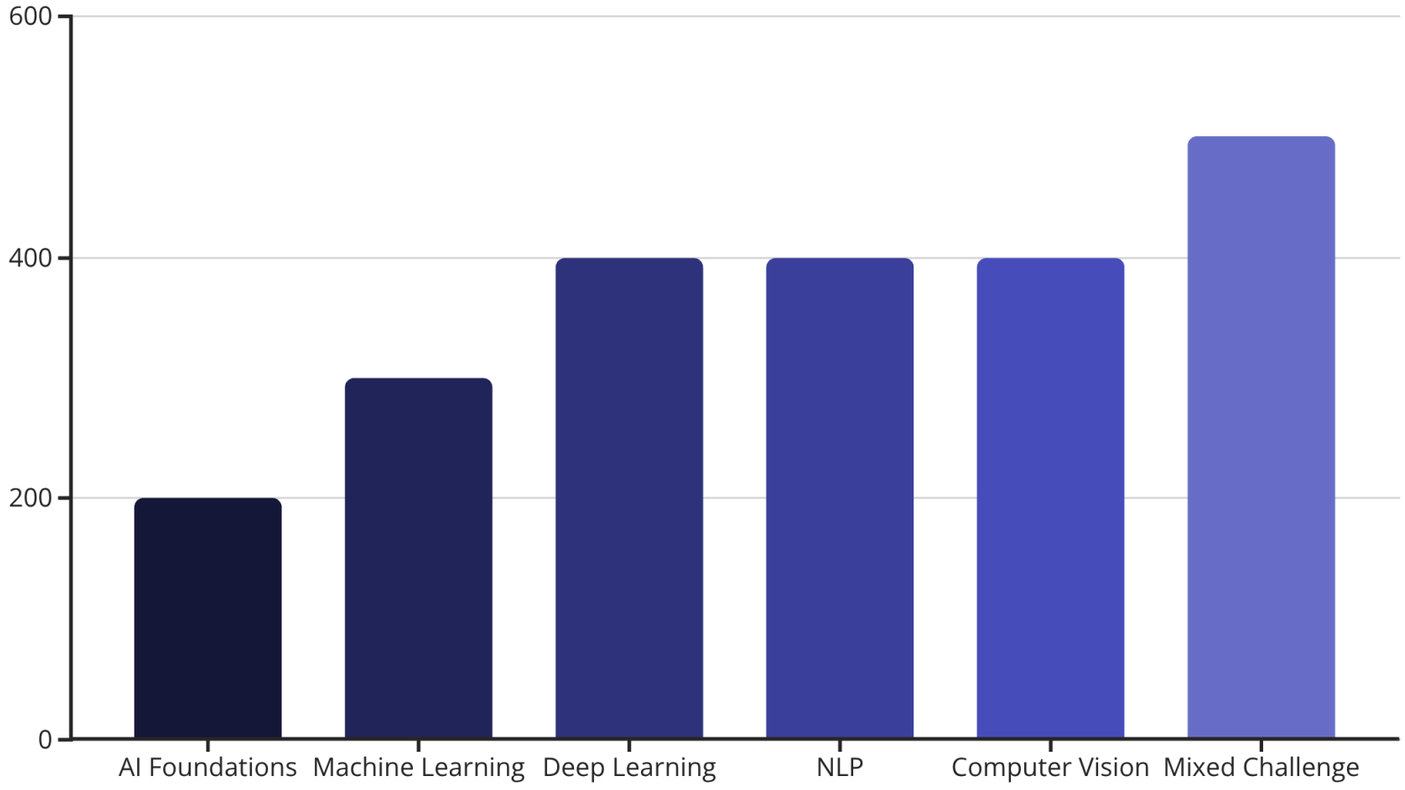


Perceptron Trainer

Adjust weights, shift the boundary, earn big. **+180 XP**

🎯 Quiz & Assessment

60+ Questions. 6 Modes. One Leaderboard.



Each mode features a **20-second timer**, streak bonuses (**+10 XP per 3-streak**), and a full results breakdown. The Mixed Challenge throws questions from all tracks — the ultimate test of your AI knowledge.

⏱️ 20s Timer

Fast-paced, high-stakes

🔥 Streaks

Chain correct answers

📊 Results

Full breakdown per mode

🔬 Real-World Simulations

6 AI Demos — Not Toy Examples

Spam Detector
Naive Bayes keyword scoring with 4 dataset presets

Sentiment Analysis
Lexicon-based NLP — pos, neg, neutral, rave presets

House Price Predictor
Linear regression with feature impact bars

Digit Recognizer
Real pixel analysis — bounding box, hole detection, softmax

Image Classifier
ResNet-50 Top-5 ImageNet results

Stock Predictor
LSTM time-series with live Canvas chart

🏆 Progress & Community

Dashboard, Leaderboard & NeuroBot

Every action earns XP. Every lesson unlocks progress. NeuroLearn keeps you motivated with a live dashboard, global leaderboard, and an AI tutor that follows you everywhere.

Live XP Dashboard

Circular progress, module bars, 7-day streak tracker

8 XP Badges + Leaderboard

From 🏆 AI Pioneer to 🌊 Deep Diver — with toast notifications

🤖 NeuroBot AI Tutor

30+ ML concepts, voice read-aloud (Web Speech API), page-aware quick chips

🛠️ Built With

Zero Dependencies. Pure Web Tech.

HTML5

Semantic markup

Vanilla CSS

Glassmorphism, dark mode

ES6+ JS

Canvas API, IIFE

Ace Editor

CDN — syntax highlighting

Skulpt

CDN — Python runtime

Web Speech API

Built-in TTS voice

localStorage

XP, auth, progress

🚀 Try It Now

No build tools. No npm. No server. Just open `index.html` in Chrome, Edge, or Firefox 90+.

✅ Clone the repo and run `start index.html` — NeuroLearn is live in seconds.

TeamVisioneers · T143 · VIBETHON"Making AI & ML education accessible, fun, and interactive for everyone." 🧠✨

Thank You!